Sumit Tapas Chongder

Master of Technology
Quantum Technologies
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Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
M.Tech QT	Indian Institute of Technology, Jodhpur	9.25 (Rank Holder)	2025-2027
B.Tech, CSE	MIT Art, Design and Technology University	8.92 (Rank Holder)	2024
Senior Secondary	Pace Junior Science College, Maharashtra	81.69%	2020
Secondary	St. Francis High School, Maharashtra	94% (Rank Holder)	2018

Experience

- **Indian Institute of Technology Jodhpur** *Jodhpur, India*
  Teaching Assistant – QCL7350: Quantum Inspired Optimization Lab *Aug 2026 – Present*
  · Teaching Executive Batch in **QCL7350 Quantum-Inspired Optimization Lab** at IIT Jodhpur, guiding hands-on work on **Max-Cut, Knapsack**, Vertex Cover, Graph Coloring, **QKP, PUBO**, adiabatic computing, and **QAOA**.
  · Supporting students in turning optimization problems into computational formulations using **Python and Qiskit**, while building lab materials, implementation guides, and practical exercises for reproducible experiments.

- **QNu Labs** *Remote*
  Summer Research Intern *May 2026 – Aug 2026*
  · Engineered and benchmarked industry-grade **CSPRNGs**, hardware **TRNGs** (microprocessors & camera sensors), and homodyne-detection **QRNGs**; validated entropy quality metrics across a **4.15 GB** multi-source binary dataset using *NIST SP 800-22*, *NIST SP 800-90B*, *Dieharder*, and *ENT* test suites.
  · Designed and deployed a hierarchical randomness source attribution pipeline extracting **64 statistical features** per window, **via** calibrated ensembles (**CatBoost, XGBoost, LightGBM**) **yielding 81.21%** mean confidence.

- **QWorld** *Remote*
  Mentor – Quantum Machine Unlearning (QIntern 2026) *Jun 2026 – Aug 2026*
  · Mentored intern(s) on **Entanglement-Weighted Pruning (EWP)** for Quantum Federated Learning, combining **QFIM sensitivity and circuit entanglement** for selective client-specific parameter removal.
  · Implemented **Qiskit VQC + FedAvg** and benchmarked five unlearning baselines, achieving 0.837 ± 0.029 **accuracy**, 0.907 ± 0.034 AUROC, and **16.4× faster unlearning** than full retraining.

- **Africa Quantum Consortium** *Remote*
  Mentor – Quantum Error Correction & Mitigation (QEM) Track *Dec 2025 – Jan 2026*
  · Mentored **5 interns** (Pan-African Quantum Challenge) on **automated QEC code construction**, **CUDA-Q** simulation, and ML decoders for **topological, stabilizer, & QLDPC codes**.
  · Co-developed a two-tier framework routing 3.3%–6.2% of surface code ($d \in \{3, 5, 7, 11\}$) syndromes to **PyMatching MWPM**, boosting accuracy to **99.81%** under **fault-tolerant circuit-level noise**.

- **Microsoft (Future Ready Talent Program)** *Hybrid*
  Software Engineer (Internship) *Feb 2023 – Apr 2023*
  · Developed responsive apps deployed via **Azure Static Web Apps** and **App Services**, integrating **Bot & Cognitive Services** to achieve 99.9% **high availability** and intelligent text analytics.
  · Built secure **Azure Storage** architectures and automated **CI/CD pipelines**, leveraging **Azure Monitor** and Application Insights to reduce system downtime by 25%.

Publications & Preprints

- **Hardware-Efficient Error Mitigation and Shot-Efficient Sampling on IBM Quantum Hardware**: Chongder, S., *arXiv:2608.28535* [quant-ph] DOI: 10.48550/arXiv.2608.28535
- **Dynamic Entanglement-Weighted Pruning for Quantum Federated Unlearning in Supply-Chain Risk Prediction**: Chongder, S., *arXiv:2608.17069* [quant-ph] DOI: 10.48550/arXiv.2608.17069
- **QR-SPPS: Quantum-Native Retail Shock Propagation and Policy Stress Simulator**: Chongder, S. T., *arXiv:2607.16275* [quant-ph] DOI: 10.48550/arXiv.2607.16275
- **Latency-Constrained Hardware-Aware Quantum Error Correction Co-Design with Adaptive Confidence-Gated Neural Decoding for the Rotated Surface Code**: Chongder, S., *arXiv:2607.05814* [quant-ph] DOI: 10.48550/arXiv.2607.05814
- **Hardware-aware Low-latency Quantum Compilation with Data-driven Lightweight Error Detection for Early Fault-Tolerant Systems**: Chongder, S. T., *arXiv:2606.07666* [quant-ph] DOI: 10.48550/arXiv.2606.07666

Intellectual Property Rights – Government of India (Copyrights/Patents)

- **Quantum Resistant Module Lattice Cryptography System** | ROC No. SW-2026022888 | *27/05/2026*
- **Analytical BB84 Quantum Key Distribution (QKD) Engine** | ROC No. SW-2026022477 | *03/03/2026*
- **Quantum Hyperspectral Imaging (Q-HSI) System** | ROC No. SW-2026022430 | *27/02/2026*
- **System And Method For Quantum Phase Space Simulation Of Quantum States** | ROC No. SW-2026023370 | *22/07/2026*
- **Quantum Native Retail Shock Propagation And Policy Stress Simulator** | ROC No. SW-2026023601 | *25/08/2026*

Projects

- **QR-SPPS: Quantum Supply Chain Risk Simulator** | *Fujitsu Quantum Challenge 2026* | GitHub
  · Engineered a 40-qubit Ising Hamiltonian on A64FX supercomputers using QARP/MPI for VQE convergence, employing ADAPT-VQE and DOS-QPE to optimize supply-chain interventions.
- **Q-OPT: Quantum Network Optimization Framework** | *IBM Quantum Hardware* | GitHub
  · Developed a security-constrained QUBO framework for adaptive QKD-PQC-QRNG network optimization, implementing QAOA in Qiskit and validating solutions on IBM Quantum hardware through parameter sweeps.
- **Spacetime Sentinels: Quantum Error Detection** | *Qiskit Fall Fest Winner* | GitHub
  · Built an interactive spacetime-code quantum error-detection game using Qiskit, implementing syndrome checks, noisy-circuit analysis, and feedback to demonstrate low-overhead quantum error detection.
- **Hardware-Aware Quantum Compilation & QED** | *IBM Quantum Hardware* | GitHub
  · Engineered hardware-aware quantum compilation combining simulated annealing, ILP mapping, XGBoost syndrome scheduling, cuQuantum acceleration, and IBM QPU validation for fault-tolerant systems under noise constraints.
- **Adaptive Surface-Code QEC Decoder** | *Quantum Error Correction* | GitHub
  · Developed adaptive surface-code decoding combining neural confidence gating with MWPM refinement, improving logical accuracy while maintaining bounded latency and CPU throughput.
- **Q-TRACE: Quantum Threat Detection** | *QML & QKD* | GitHub
  · Developed physics-informed quantum machine learning for zero-day threat detection in practical decoy-state QKD, benchmarking six attacks against classical baselines and IBM Quantum hardware.

Skills

Programming Languages	Python, CUDA-Q, Rust, C++, SQL
Quantum Frameworks	Qiskit, PennyLane, Strawberry Fields, QuTiP, Cirq
ML Libraries	TensorFlow, PyTorch, Scikit-learn, Keras, NumPy, Pandas, Matplotlib
Cloud & DevOps	Microsoft Azure (App Services, Static Web Apps, Storage)
Other Tools	Git, GitHub, Streamlit, VS Code, MySQL, Jupyter Notebook

Licenses & Certifications

- **Qiskit Global Summer School 2026 - Quantum Excellence (QGSS26 Mentor)** – IBM | Credential Link
- **Foundations of Quantum Error Correction** – IBM | *Jun 2026* | Credential Link
- **Integrating Quantum and High-Performance Computing** – IBM | *Jun 2026* | Credential Link
- **Quantum Machine Learning** – IBM | *Jun 2026* | Credential Link
- **Practical Introduction to Quantum-Safe Cryptography** – IBM | *Jun 2026* | Credential Link
- **QCourse 581-1: Selected Topics in Quantum Science and Technologies** – QWorld | Credential Link
- **QSilver: Diploma in Quantum Computing & Programming** – QWorld | Credential Link

Honors, Awards & Hackathons

- **1st Place – IBM Qiskit Fall Fest IIT Jodhpur Hackathon 2026** | IBM Quantum & IIT Jodhpur | *Oct 2026*
  Project: "Spacetime Sentinels: Gamified Quantum Error Detection" – Developed an interactive spacetime-code-based quantum error detection game using Qiskit with syndrome checks and noise analysis. | GitHub | Certificate Link
- **1st Place – IBM Qiskit Fall Fest IIT Jodhpur Hackathon 2025** | IBM Quantum & IIT Jodhpur | *Nov 2025*
  Project: "AnalytiQ: Analytical BB84 QKD Engine" – Implemented and simulated QKD protocols with focus on security, channel integrity, and quantum communication workflows using Qiskit. | GitHub | Certificate Link
- **Top 50 – MIT iQuHack 2026 (Rank 46 Worldwide)** | BlueQubit X MIT | *Jan 2026* | Certificate Link
- **Top 40 – Yale Peaked Hackathon 2026 (Rank 38 Worldwide)** | BlueQubit X Yale | *Apr 2026*

Leadership & Volunteering

- **Qiskit Advocate** – **IBM Quantum**: Selected for global program; contributing to quantum computing community.
- **Campus Ambassador** – Talent Battle (May 2023 – Nov 2023): Led workshops and events; increased student participation through campus outreach.
- **Student Volunteer** – HelpAge India & Indian Red Cross Society: Contributed to community service initiatives.